

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)	I	63 = 3 marks If incorrect award 2 marks for 63.4 If incorrect award 1 mark for $\frac{317}{5}$ or $\frac{62+65+67+60+63}{5}$		3		3	3	3
			II	Girls (heart) (rate/ bpm/ mean) is {higher/ faster/ more} (than boys)/ ORA Ignore bigger/ smaller with reference to rate ecf from (I) ignore ref to fitness		1		1		1
		(ii)		Any one (×1) from: picked at random (1) all sat (at rest) for {one minute/same amount of time} (1) same age/ class (1) same (smart)watch (1) Ignore same numbers of boys and girls / ref to fitness			1	1		1
	(b)	(i)	I	All 5 plots correct = 2 3 or 4 correct = 1 Tolerance +/- less than one small square		2		2	2	
			II	accurate line Tolerance +/- less than one small square		1		1	1	
		(ii)		{Yes/data do support/ owtte} + women have higher <u>rate</u> (than men) (ORA) ecf from (i)I			1	1		

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		(iii)		Rises, then falls (1) from 64.5 (bpm)/(age) 50 (1) peaks at {64.5 (bpm)/50 (yrs of age)} = 2 marks ecf from graph for peak value		2		2		
	(c)			Any two (×1) from: Thousands of people/more data/more results/ more people/ bigger sample size (1) Many nations/ from all around the world (1) Many ethnicities/more diverse (1) More ages (1) carried out over longer period of time (1) Note: Reverse arguments apply			2	2		2
				Question 4 total	0	9	4	13	6	7